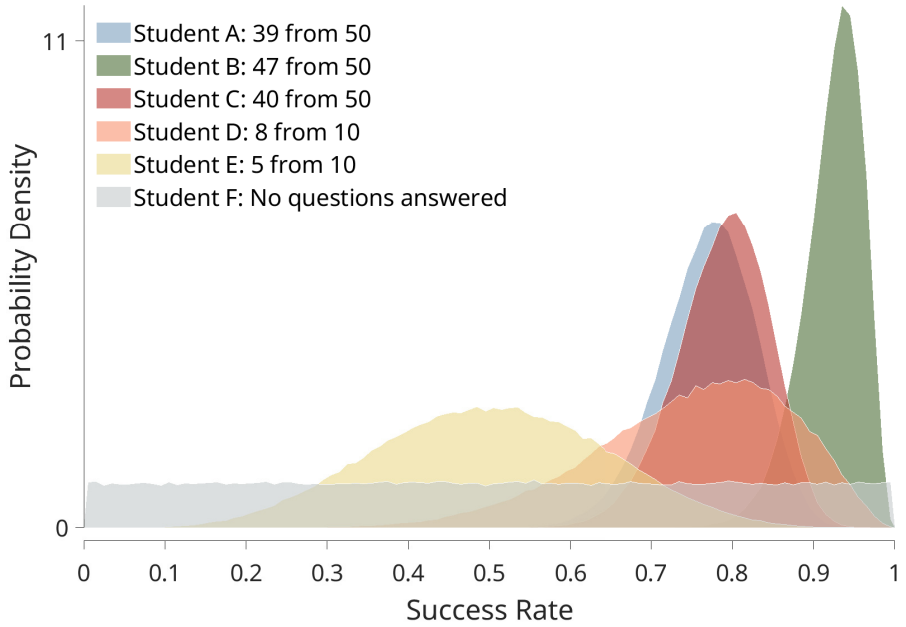
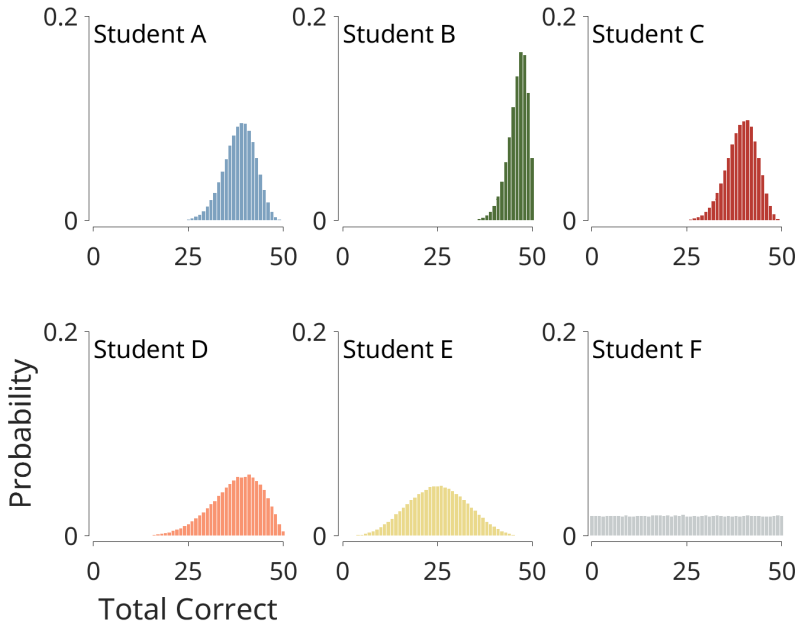
Hierarchical Success Rate Model

Michael D. Lee

Intuitions

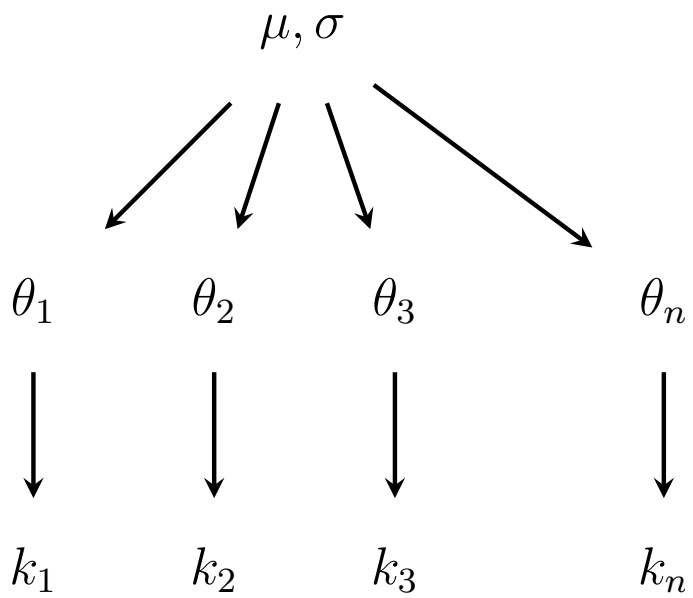
- Do the posterior inferences for students E and F seem the most reasonable? Does the prediction for student F seem the most reasonable?
  - it seems conceivable, even likely, that student E just got off to a bad start, and will end up having a grade above 50% correct
  - even though student F has not answered any questions, it seems likely, for example, their accuracy rate will be above 0.5 rather than below 0.5, and they will likely also score better than 50%





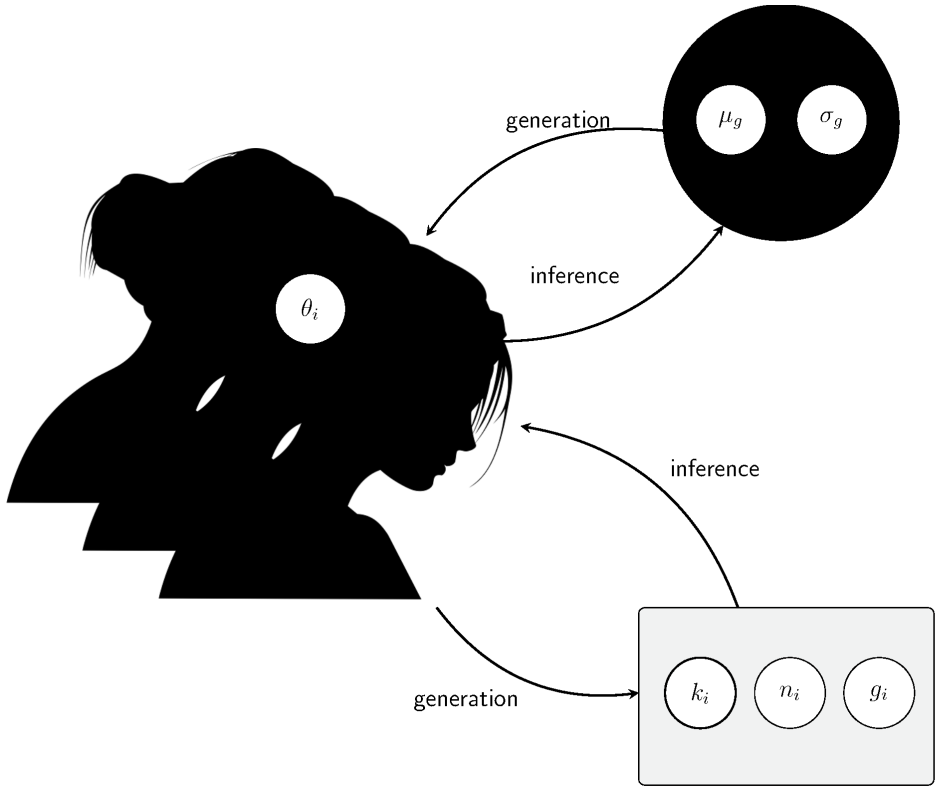
Hierarchical Model

- The mismatch between intuition, and the model's inferences and predictions, arise because of the assumption that students are independent
- Hierarchical models allow both sameness and difference to be modeled, by assuming individual-level parameters that are connected by all being drawn from an over-arching group distribution
  - for this example, the hierarchical distribution is the class curve



Hierarchical Success Rate Model Structure

- The hierarchical success rate model has the structure
  - observed data k_i successes out of n_i observations
  - known allocation of each individual to a group g_i
  - latent parameters for the success rate of each individual θ_i
  - the success rates come from Gaussian group distributions with means μ_g and standard deviations σ_g



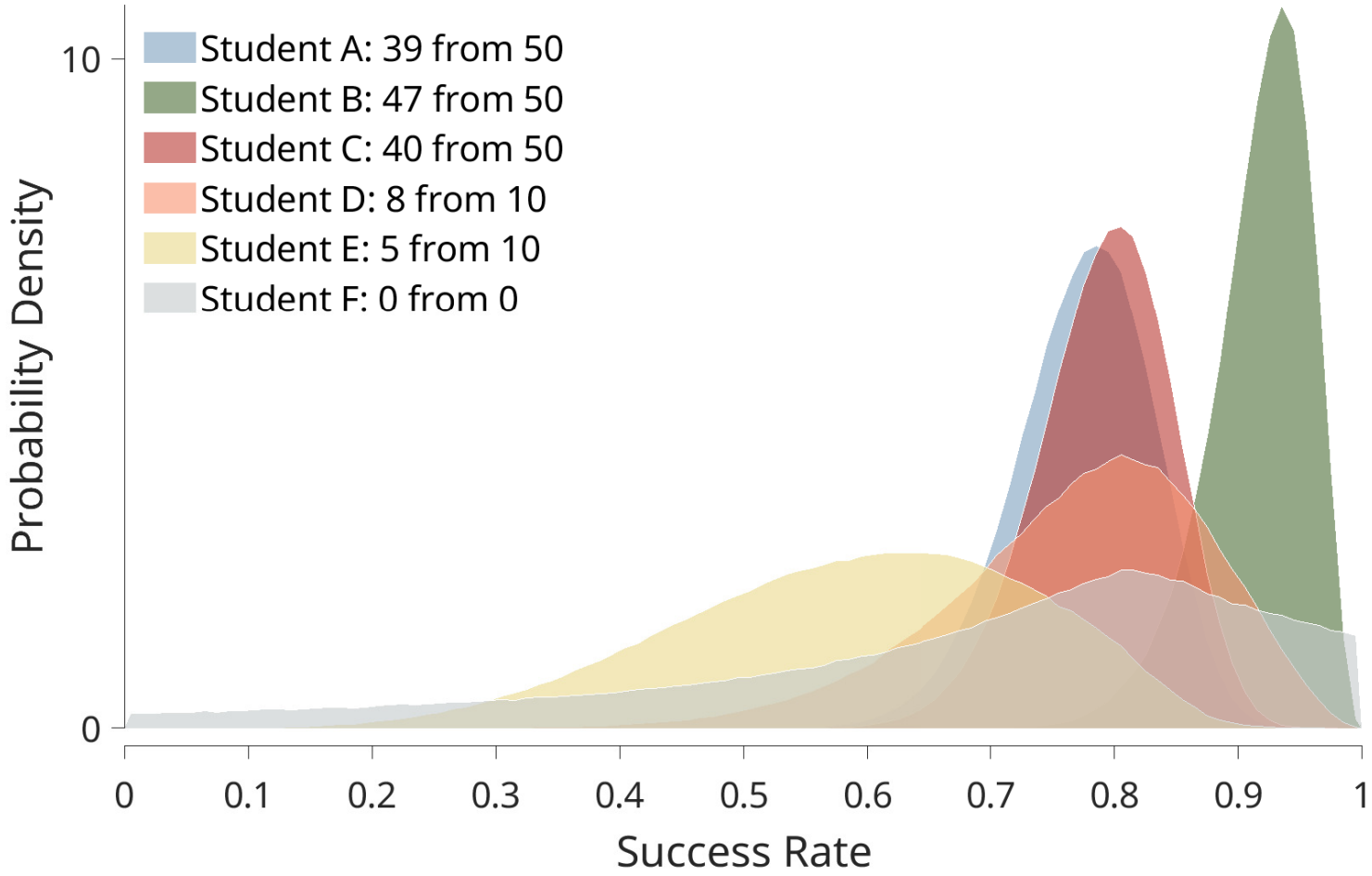
Hierarchical Grades

Hierarchical Grades

- The students are now hierarchically related to each other, through their shared membership of a single group distribution
  - i.e., in this application, all $g_i = 1$ and there is a single μ and σ
- Hierarchical models allow inferences at multiple levels of abstraction
  - inferences about μ and σ are inferences about the class curve
  - inferences about θ_i are inferences about individual students
- Hierarchical models of individual differences can capture both what people have in common and how they differ from one another

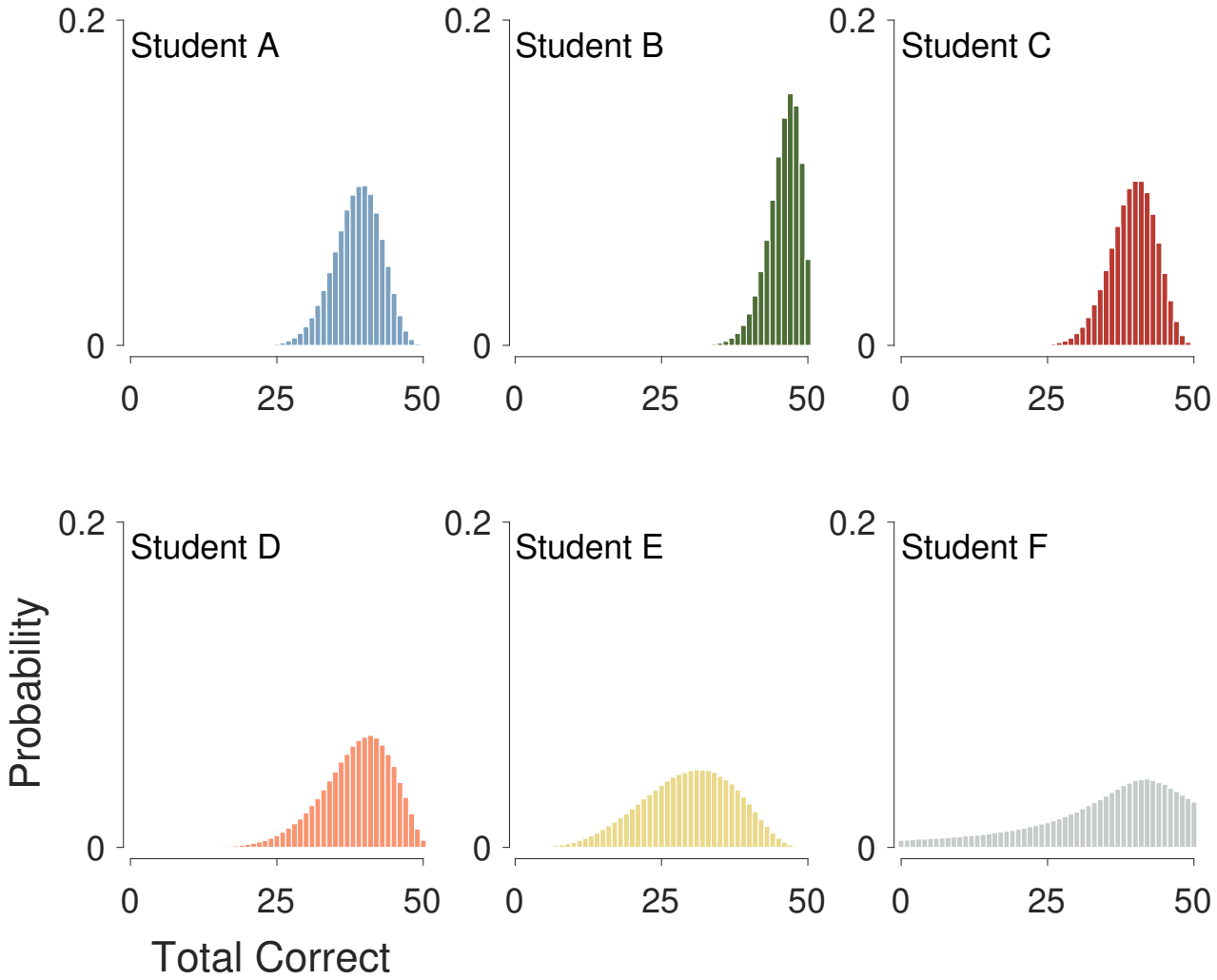
Individual Success Rate Posteriors

- The posteriors for θ_i based on the hierarchical model are shown
  - the inferred success rates for all students, but noticeably students E and F, are influenced by the group distribution by an effect called "shrinkage" (or "sharing statistical strength")



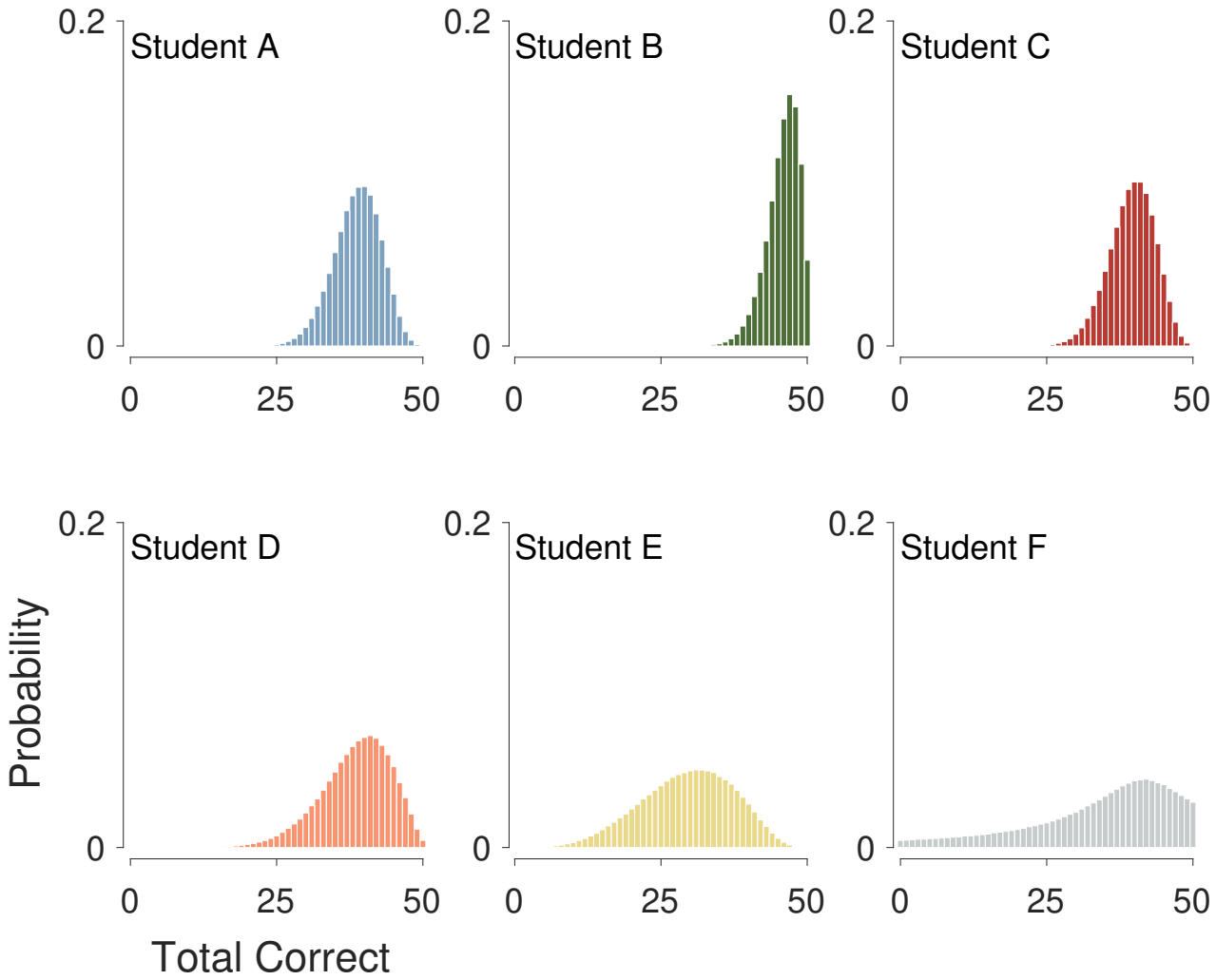
Individual Success Count Posterior Predictions

- The posteriors predictive distributions $\hat{k}_i$ for $\hat{n} = 50$ are shown
  - students A, B, and C: little change from the non-hierarchical model



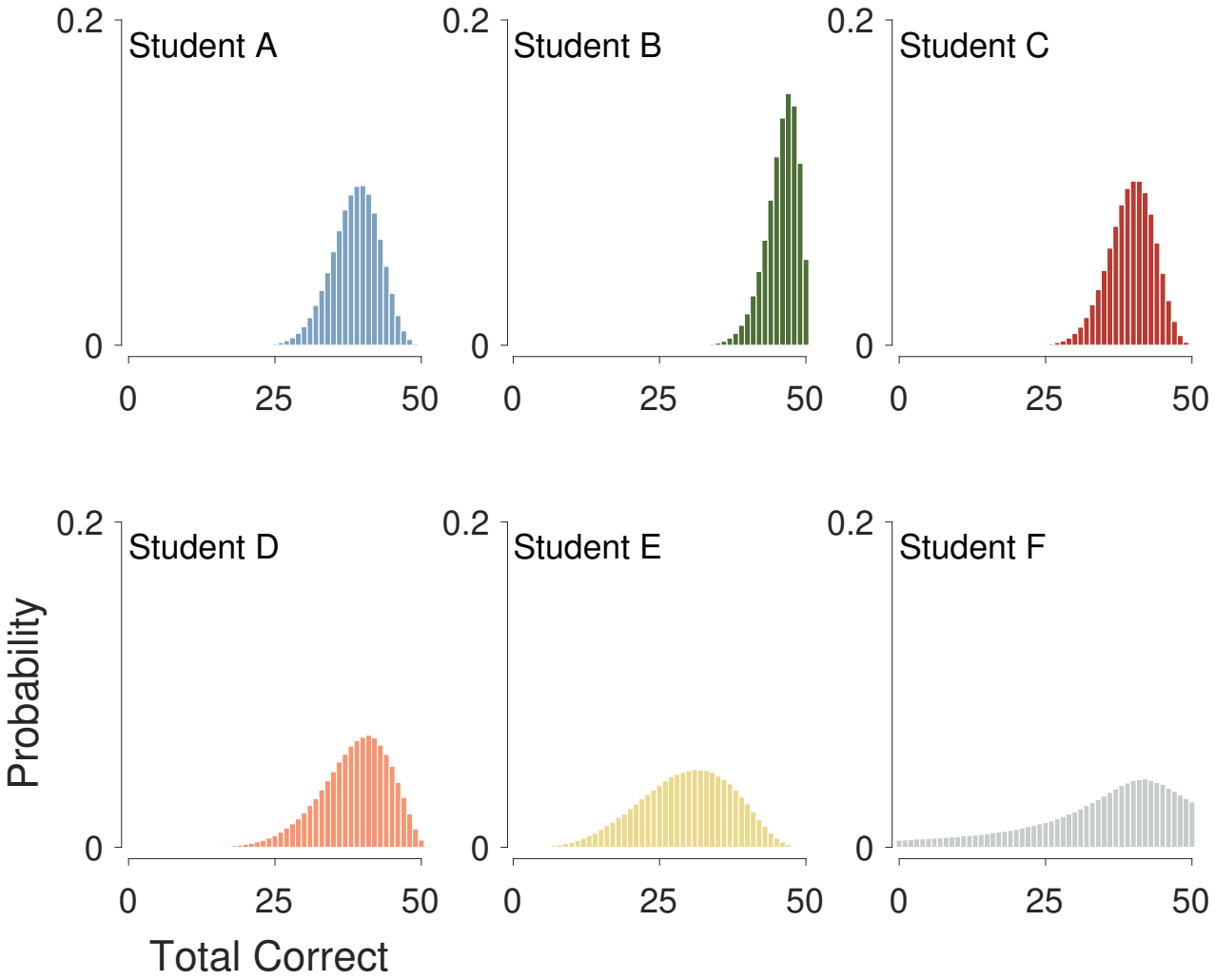
Individual Success Count Posterior Predictions

- The posteriors predictive distributions $\hat{k}_i$ for $\hat{n} = 50$ are shown
  - students D and E: predict more correct answers



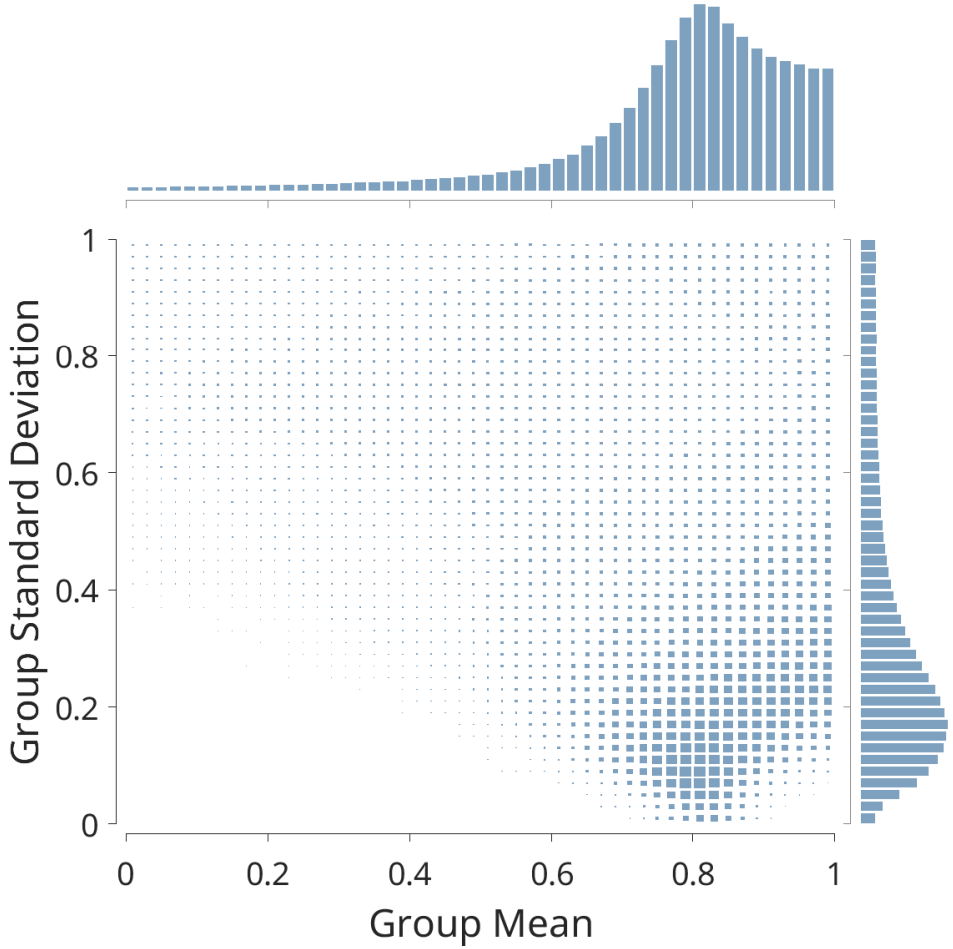
Individual Success Count Posterior Predictions

- The posteriors predictive distributions $\hat{k}_i$ for $\hat{n} = 50$ are shown
  - student F: corresponds to a representative prediction k for a "random student" or "the next student"



Group Mean and Standard Deviation

- The joint and marginal posterior distributions for the group-level μ and σ parameters are shown
  - the class curve could have a mean of 80% with little variability, or a higher mean with more variability



Key Points

- Hierarchical models extend a basic cognitive model to include an account of how the parameters themselves are generated
  - different inferences: success rates are "sandwiched" between the group distribution that generates them and the data they explain
  - new inferences: the properties of the group distribution are inferred, as well as inferences about a representative individual
- Hierarchical models make different inferences and predictions, because they make different assumptions
  - whether the hierarchical model inferences are better or worse depends on the usefulness of the assumptions

